

RESOURCE ARTICLE

Beyond Presence and Absence: Using eDNA and Microsatellite Genotyping to Estimate Densities of Microscopic Life Forms in Wild Populations

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ABSTRACT

Many challenges arise when monitoring organisms with cryptic life-histories. For example, some cryptic life-stages are hard to identify or sample due to their microscopic nature, which creates unknowns surrounding an organism's population dynamics. Environmental DNA (eDNA) is a non-invasive sampling technique used to monitor cryptic species when traditional survey methods are challenging. Generally, eDNA has been used to quantify the presence/absence of species in various habitats. However, recent advances in high-throughput amplicon sequencing techniques have enabled researchers to detect intraspecific genetic diversity with eDNA. In this study, we present two complementary R packages that can be used to estimate the number of individuals in an eDNA sample. The first package (Amplicomsat) cleans high-throughput amplicon microsatellite sequences and counts the observed alleles identified in eDNA. Our second package (GenotypeQuant) then uses a numerical maximum likelihood estimator (NMLE) to estimate the number of contributors most likely to have produced the sequenced panel of microsatellite alleles amplified from eDNA. We first present simulations to characterise the accuracy and precision of the method. We then estimated densities of *Nereocystis luetkeana* (bull kelp) microscopic gametophytes from eDNA collected from an experiment with a manipulated number of gametophytes. Finally, we analysed benthic eDNA from kelp forest habitats. We found that gametophyte estimates produced by the NMLE varied within +3/−2 individuals when processing eDNA from rocks with 8 seeded gametophytes. We estimated 500 to 800 gametophytes·m^{−2} densities in July, five or more months since spore germination and before the current year's spore release. Gametophyte abundance scaled with the sampling area and numbers were higher than total sporophyte densities.

1 | Introduction

Environmental DNA (eDNA) extracted from water or sediment is a common non-invasive and cost-effective approach to monitoring aquatic organisms (Jerde et al. 2011; Sigsgaard et al. 2020). Through the collection of cells, hair, skin, and faeces from an environmental sample, eDNA methodologies allow for the capture of genetic material without directly sampling tissue from an organism (Taberlet et al. 2018). eDNA analysis is currently being used to estimate species-level biodiversity through presence/absence (Ficetola et al. 2008) and relative abundances (Stoeckle et al. 2021) when other forms of sampling efforts are challenging, costly, or impossible to complete (Fernández et al. 2018; Stat et al. 2019; Adams et al. 2019). For example, eDNA techniques are a useful tool for detecting the presence of endangered, transient, deep, and cryptic species when densities are low and impossible to detect using traditional field survey methods (Jerde et al. 2011; Port et al. 2016; Stoeckle et al. 2021). With advancements in next-generation sequencing methods, molecular monitoring through the collection of eDNA provides a platform that can improve our understanding of species distributions and composition. These advancements are being broadly applied to inform habitat conservation and management practices (Bohmann et al. 2014; Beng and Corlett 2020; Sahu et al. 2023; Acharya-Patel et al. 2024).

While eDNA has been extensively used for taxonomic biodiversity monitoring, researchers are developing tools for obtaining intraspecific abundance by optimising sequencing technologies to estimate species' biomass and abundance from an environmental sample (Baker et al. 2018; Andres et al. 2021, 2023; Suarez-Bregua et al. 2022). For example, studies have used eDNA to detect different haplotypes of the same species. However, the same haplotype sequenced from a mixed sample could have come from one or many individuals (Baker et al. 2018; Suarez-Bregua et al. 2022). This uncertainty has led to the development of techniques such as quantitative polymerase chain reaction (qPCR), which detects the number of target sequence copies for a species of interest. Additionally, sequence reads obtained from metabarcoding techniques can be used to estimate relative read abundances for species identified in an eDNA sample. Both qPCR and relative read abundance indirectly estimate species-specific densities by assuming that species abundance scales with the number of reads or available DNA template (Spear et al. 2021; Stoeckle et al. 2021). Though these methods have been demonstrated for various fish species (e.g., *Sander vitreus* and *Sebastes* spp.), the relationship between gene copies in an environmental sample and densities for other species, including non-motile species, is understudied (Acharya-Patel et al. 2024; Blackman et al. 2024).

Though eDNA is generally used in qPCR and metabarcoding, DNA extracted from an environmental sample can be used to estimate the number of individuals in a sample using microsatellite markers (Weir et al. 1997; Adams et al. 2019; Sethi et al. 2019). Microsatellites are co-dominant genetic markers commonly used to fingerprint individual samples, resulting in multilocus genotypes where each locus can have a maximum number of alleles equal to the organism's ploidy

level. In forensic analysis, when the goal is to estimate the number of individual genetic contributors in a mixed sample (hereafter, number of contributors, NOC), the number of microsatellite alleles found per locus can be higher than the ploidy, resulting in mixed allele profiles that are impossible to interpret (Clayton et al. 1998). These observed allele counts (OAC) over multiple loci can be used to estimate the NOC in the mixed sample using maximum likelihood approaches (Weir et al. 1997; Curran et al. 1999; Haned et al. 2011; Sethi et al. 2019).

Originally, forensic mixture analyses were designed with a strong assumption of known NOCs— the victim, suspects with alibis, and suspects of interest. This limitation produced downstream estimation biases, given that DNA contributions from a single individual can be small or masked by signals of other contributors (Haned et al. 2011; Bright et al. 2014; Benschop et al. 2015). These models have been adapted to use available allele frequencies of the target population to derive likelihoods, removing the need for a priori NOC assumption (Egeland et al. 2003). This approach has enabled the use of mixture models beyond forensic analyses, extending its application to ecological modelling (Sethi et al. 2019; Andres et al. 2021, 2023). While collecting an environmental sample, several individuals of the same species are likely to be captured; therefore, the framework to estimate NOC in forensic analysis has been naturally adapted to biological studies. The R package *forensim* (R Core Team 2024) implements the maximum likelihood approach in Curran et al. (1999) but can only estimate a maximum NOC of six. Sethi et al. (2019) used a more computationally efficient approach to estimate yellow perch's NOC in mixed samples from stomach contents of largemouth bass. Recently, Andres et al. (2023) characterised the precision and accuracy of Sethi's approach under different scenarios to solve simulated mixtures up to a NOC of 100. This capacity is a welcomed addition to the molecular ecologist toolbox, given that high NOCs can be found in eDNA samples from various biological contexts.

Currently, ecological applications using forensic mixture analyses require two complementary datasets—sequenced eDNA, from which the OAC for each microsatellite locus is derived, and the population allele frequencies for the study region (Sethi et al. 2019; Andres et al. 2021, 2023). Though many improvements have been made to this approach over the years, a few assumptions remain. For example, using a reference allele frequency pool can only be effective when the population of interest does not deviate from Hardy–Weinberg equilibrium (Egeland et al. 2003). This assumption in the method would produce estimation biases when studying species with selfing and mixed mating systems that deviate from the probabilities expected under panmixia. While these challenges make analytical maximum likelihood solutions more complex, numerical approaches can be explored to address these issues. Moreover, in the empirical implementation of these methods, due to PCR and sequencing shortcomings, some alleles in a mixed sample might fail to amplify (allele dropout) or be scored ambiguously (allele stutter). Current high-throughput sequencing (HTS) of microsatellite amplicons alleviates these concerns while allowing for increased multiplexing of samples and markers (Lepais et al. 2020).

Amplicon sequencing will also reveal alleles that differ in sequence polymorphism, which would otherwise exhibit homoplasia with sequence size. Detecting these sequence-based alleles increases the number of rare alleles detected, which has been shown to improve the precision of the NOC estimates (Andres et al. 2023).

eDNA methods to estimate NOCs in mixed samples can be a tool to monitor the population dynamics of species with complex life histories. Many marine organisms (e.g., sea stars, urchins, macroalgae) have a microscopic life phase that minimises competition between parents and offspring (Marshall and Morgan 2011). For example, the bull kelp (*Nereocystis luetkeana*) has a bi-phasic annual life history, where the large forest-forming diploid sporophyte phase alternates with the microscopic haploid gametophyte phase (Schiel and Foster 2006). Bull kelp sporophytes release spores (in summer and early fall) prior to senescence (Rigg 1912; Springer et al. 2010). Spores then germinate into gametophytes which live on the benthos during winter. Kelp gametophytes survive during winter, delaying reproduction and corresponding sporophyte recruitment when growth conditions are optimal during spring and summer (Carney and Edwards 2006). Both stages are self-sufficient and free-living, achieving multicellularity and interacting with their surrounding environment in different ways (Schiel and Foster 2006). Thus, the ability of both stages to tolerate disturbances is important in keeping kelp populations healthy, stable, and abundant. Long-lived gametophytes may create microscopic banks that are analogous to terrestrial seed banks, which supply multigenerational genetic diversity to sporophyte recruits (Carney and Edwards 2006; Carney 2011; Carney et al. 2013). However, there is little known about the population dynamics of these gametophyte banks, e.g., the fluctuations in their densities and life expectancy—as it is extremely difficult to find and monitor this cryptic life stage in situ. Bull kelp forms subtidal temperate forests that create uniquely biodiverse habitats along the coastline of the Northeast Pacific (Arafah-Dalmau et al. 2023). However, the abundance of these kelp forests has been regionally declining, raising concerns over losses of entire ecosystems (Pfister et al. 2018; Berry et al. 2021; Arafah-Dalmau et al. 2023; Cavanaugh et al. 2023). Specifically, increased grazing pressure from sea urchins has caused extreme shifts from healthy kelp forests to urchin-dominated ecosystems (urchin barrens; Filbee-Dexter and Scheibling 2014; Rogers-Bennett and Catton 2019), with some communities unable to recover fully (Burt et al. 2018; Hind et al. 2019).

In this study, we build upon previous eDNA mixture analysis methods to quantify the densities of microscopic life forms. In particular, we developed an R package (AmplicomSat) to score HTS amplicon sequence data and output the OAC present in an eDNA sample. We also developed an R package (GenotypeQuant) that uses numerical maximum likelihood (hereafter NMLE) to estimate NOC in an eDNA sample. Additionally, this package includes quality control analyses, allowing the user to detect estimation biases caused by PCR effects, DNA quality, or deviations from Hardy–Weinberg equilibrium. We present simulation studies on the precision and accuracy of the method. We then use the approach to quantify bull kelp's microscopic gametophyte life-stage in a lab

experiment and field study. First, we used a controlled seeding experiment where numbers of gametophytes are known a priori, which allowed us to test the effect of allele richness (AR) and evenness. Second, an eDNA collection from benthic rock samples in kelp forest habitats for which we hypothesise that estimated gametophyte densities from wild kelp forests will scale with the rock sampling area. These tests ultimately demonstrate the effectiveness of the NMLE in counting individuals mixed in an environmental sample, bringing the utilisation of eDNA further beyond presence–absence.

2 | Materials and Methods

2.1 | Microsatellite Characterisation

We used QDD (Megléc et al. 2010), which integrates PRIMER 3 (Untergasser et al. 2012), BLAST (Altschul et al. 1990), and CLUSTALw2 (Larkin et al. 2007) to design microsatellite markers from a bull kelp reference genome (assembly: uoNerLuet1.0.; GenBank: GCA_031213475.1). We followed QDD parameterization optimised for amplicon sequencing (Lepais et al. 2020). We selected loci with tetranucleotide motifs for ease of scoring due to reduced levels of stutter. We tested 50 loci for amplification and polymorphism. A total of 36 loci produced amplification while 23 were considered to be polymorphic. The genotyping employed in this study used 10 of these loci (Table S1) in addition to a locus from Gierke et al. (2023; Ner 2), making a total of 11 markers (Table S1).

2.2 | Validating Techniques for Amplifying Microsatellites in an eDNA Sample

We initially tested if all alleles present in a mixed sample could be scored when AR and evenness vary. Gametophyte cultures ($n = 22$) were genotyped separately from the University of Wisconsin-Milwaukee bull kelp biobank for three markers (NL26, NL27, Table S1; Ner2 from Gierke et al. 2023), using the Phire Plant Direct Kit (ThermoFisher Scientific). Live tissue from each culture was harvested and suspended in 20 μ L of Phire dilution buffer. Each sample was vortexed for 10 s and incubated at room temperature for 5 min. 0.5 μ L of the solution was used as a DNA template in a 20 μ L PCR, containing 10 μ L of 2x Phire Plant PCR buffer, 0.4 μ L Phire Hot Start II DNA polymerase, 8.8 μ L water, 5 μ M of 5' fluorochrome labelled forward primer, and 10 μ M of reverse primer. The PCR protocol consisted of an initial denaturation at 98°C for five minutes followed by 40 cycles of 98°C for 5 s, 58°C for annealing for 5 s, and 72°C extension for 20 s. Samples were treated with an additional extension step at 72°C for 20 min after cycling. PCR product fragment sizes were separated and scored on an ABI 3730 (Applied Biosystems) using GeneScan-500 LIZ as the size standard (at the University of Illinois Roy J. Carver Biotechnology Center). Alleles were scored using STRAND v. 2.4.110 and binned using the MsatAllele v. 1.0.4 R package (Alberto et al. 2009). Extracted gametophyte DNA from the genotyped cultures was then homogenised and mixed in a 96-well plate, simulating eDNA samples (Table S2) that varied in AR (2–5 alleles) and evenness (1–0.1). DNA from the

mixed samples was genotyped and scored using the same methods detailed above.

2.3 | Numerical Maximum Likelihood Estimator (NMLE)

Under a maximum likelihood framework, the observed count of unique alleles amplified at each locus (observed allele count, OAC) can be used to estimate the most likely number of genotypes (here gametophytes) that could have contributed to the field sample. Below the general likelihood is calculated in Equation (1):

$$L(\theta; OAC) = \prod_{l=1}^n f(oac_l | \theta) \quad (1)$$

with oac_l the count of observed unique alleles at locus l , OAC the vector of oac values across all loci, and parameter θ the number of haploid genotype contributors (NOC) that originated them. In our numerical approach, $f(oac_l | \theta)$ terms are derived for each locus l from sampling distributions of random variable oac_l , with different distributions for each value of the θ parametric space, from $\theta = 1$ to $\theta = \max \theta$, with maximum θ defined by the user. These sampling distributions are generated from known allele frequencies of a reference population. As such, $f(oac_l | \theta)$ is the proportion of times the observed allele count at locus l was sampled in each distribution generated for each θ value. The estimated value of θ for locus l is the one with the highest proportion of sampled values equal to oac_l (maximises function $f(oac_l | \theta)$). The multilocus estimate of the NOC is obtained by taking the log of $f(oac_l | \theta)$, summing across loci, and finding the θ value that corresponds to the maximum of this log-likelihood function.

We developed an R (R Core Team 2024) package, GenotypeQuant (<https://github.com/UWMAAlberto-Lab/GenotypeQuant.git>), to implement and distribute the NMLE of NOC given the OACs scored from an eDNA sample. GenotypeQuant estimates an approximate 95% confidence interval with a likelihood ratio approach, where the confidence interval includes all θ (gametophyte) values for which the log-likelihood function drops off by no more than 1.92 units. GenotypeQuant also provides an option to correct for any deviations of Hardy–Weinberg equilibrium if the reference population has high values for F_{is} . The correction for F_{is} is incorporated by changing how the sampling distributions of oac_l are sampled from the reference allele frequencies. First, the R package Genepop (Rousset 2008) is used to calculate Q_{intra} , the observed frequencies of identical pairs of genes within individuals. Alleles are then sampled sequentially with probability Q_{intra} to sample the previously sampled allele and $1 - Q_{intra}$ to sample any other allele according to their allelic frequencies.

When mixtures contain high NOCs, loci may saturate when all alleles in the reference population are scored in a sample. When all possible alleles are observed at a given locus, the likelihood that a certain NOC produces the observed set of alleles is positively biased (Egeland et al. 2003). To address this problem, GenotypeQuant provides an additional

corrected estimate by removing ‘saturated’ loci from the log-likelihood function, using similar methods reported by Andres et al. (2023).

Lower quality yields of DNA extracted from an environmental sample may be missing fragments associated with a particular microsatellite marker, which results in a sample that fails to amplify all the microsatellite alleles present. More concerning, for problematic samples, there might be heterogeneity across loci in the number of alleles that are amplified. This can result in observed allele counts at some loci that deviate from those expected, given the AR of each locus in the reference population. For example, a marker with more alleles in the reference population might have fewer amplified alleles in the eDNA sample than a locus with fewer alleles in the reference population. This will result in decreased accuracy and non-informative confidence intervals. We used a Chi-square statistic to measure the deviation between the observed and expected number of alleles at each locus, given the diversity of the different loci in the reference population and the total number of alleles amplified. The Chi-square statistic can be seen as a flag measuring the drop in the quality of the eDNA sample, allowing users to exclude such loci from the analysis.

2.4 | Testing Precision and Accuracy of the NMLE

We used simulation to study the precision of the NMLE as a function of the polymorphism in the genotyping panel, measured by the number of loci used and their maximum AR. Genotype data was simulated to match different combinations of loci number and AR using the function *sim.genot* from the hierfast R package (Goudet 2005). We then sampled simulated eDNA from each simulated population of a given combination of number of loci and AR. Each sample contained a number of haploid genotypes (NOC) that varied from 1 to 50, with 50 replicated samples per NOC value. We then calculated the mean precision by averaging over the replicated samples the width of the 95% CI for the NOC estimate produced by GenotypeQuant, for all combinations of number of loci and AR explored.

The size of the reference population used when estimating NOC determines the chance of sampling rare alleles, which have been shown to reduce negative bias (Sethi et al. 2019; Andres et al. 2023). We used a simulation to study the precision and accuracy of the NMLE estimate as a function of reference population size. We compare our results with an analytical maximum likelihood approach used in Andres et al. (2023), testing both methods with and without a correction for ‘saturated’ loci. We added GenotypeQuant’s NMLE approach to an edited version of the code used in figure 2 of Andres et al. (2023), which used a large data set of linesnout gobies (*Elacatinus lori*) microsatellite genotypes (D’Aloia et al. 2020). We sampled genotypes to create simulated mixed samples with NOC ranging from 2 to 70, with 25 replicates, and ran the different NOC estimators. We compared three levels of reference population size: 25, 40, and 100. We kept the number of loci at 10 to simulate a number closer to what is likely to be used in most studies. We measured how accuracy, quantified as bias (real—estimated NOC), and precision, quantified as the standard deviation of the NOC estimate among replicates,

were influenced by the real NOC in the sample, the size of the reference population, and the estimator used.

2.5 | Method Validation With Experimental Mixed Samples of Known NOC

Before utilising the method to estimate NOC from eDNA samples collected in the wild, we validated the NMLE method using mixtures of gametophytes with known genotypes. During the summer of 2022, we collected reproductive kelp sori ($n=40$ blades) from different individuals growing in Owen Bay, British Columbia Canada (50.307, -125.227). Kelp spores were released and settled in petri dishes for gametophyte propagation and cultivation (Littler et al. 1973). After one month in culture conditions, we identified and isolated single-genotype gametophyte cultures (Luthringer et al. 2014). Once isolated, we maintained the gametophyte cultures to grow and increase in size (Littler et al. 1973).

Eight gametophyte cultures from the Owen Bay collection were used to create artificial gametophyte banks. These cultures were used to test if the NMLE could accurately estimate gametophyte numbers from an eDNA sample, using the gametophyte genotypes as our known pool of alleles. Gametophytes from each culture were fragmented in a blender and vacuum-filtered to remove excess media. Once filtered, all fragments were pooled and resuspended in 200 mL of sterile seawater media (natural, filtered and autoclaved seawater with an f/2 supplement). Pooled gametophyte fragments (2.5 mL) were mixed with 2.5 mL of food-grade sodium alginate solution (20 g/L) and painted onto settlement rocks ($n=16$; Figure 1 B–D). The painted rocks were sprayed with a calcium chloride solution and left to cure for 20 min. Once seeded and cured, the rocks were incubated at 12°C in flow-through tanks with 1 μ m filtered and UV sterilised seawater, red light, and 12:12 light/dark daylengths for 18 days. This incubation period allowed the gametophytes to settle and grow prior to eDNA sampling (Figure 1B,D).

After the allotted growth period, each rock was sprayed clean with sterile seawater and scrubbed with a sterile toothbrush. The liquid waste from the rock became the eDNA sample used to detect gametophytes. The liquid samples were filtered with 0.22 μ m Sterivex DNA filters (Millipore Sigma), after which each filter was fixed in Longmire's solution and stored at 4°C. Seawater was sampled for eDNA from each tank to determine if kelp detritus was coming into the tanks via the flow-through system and thus compromising the sequencing of seeded gametophytes. In addition, three negative control samples were collected to test the presence of kelp detritus in the sterilised seawater used to rinse our rock samples. The sterile seawater was filtered with a 0.22 μ m Sterivex DNA filter and fixed with Longmire's solution.

eDNA samples were extracted using a modified version of the NucleoSpin 96 Plant Kit II (Macherey-Nagel, Duren, Germany), where 800 μ L of lysis buffer and 20 μ L of RNase A were added directly into the Sterivex filter along with 0.3 g of fine silica beads. The Sterivex filter was vortexed for 10 min and incubated at 65°C for 1 h. The lysate was then extracted from the filter using a sterile syringe, after which we proceeded with the recommended extraction steps. All reusable equipment, including filter caps,

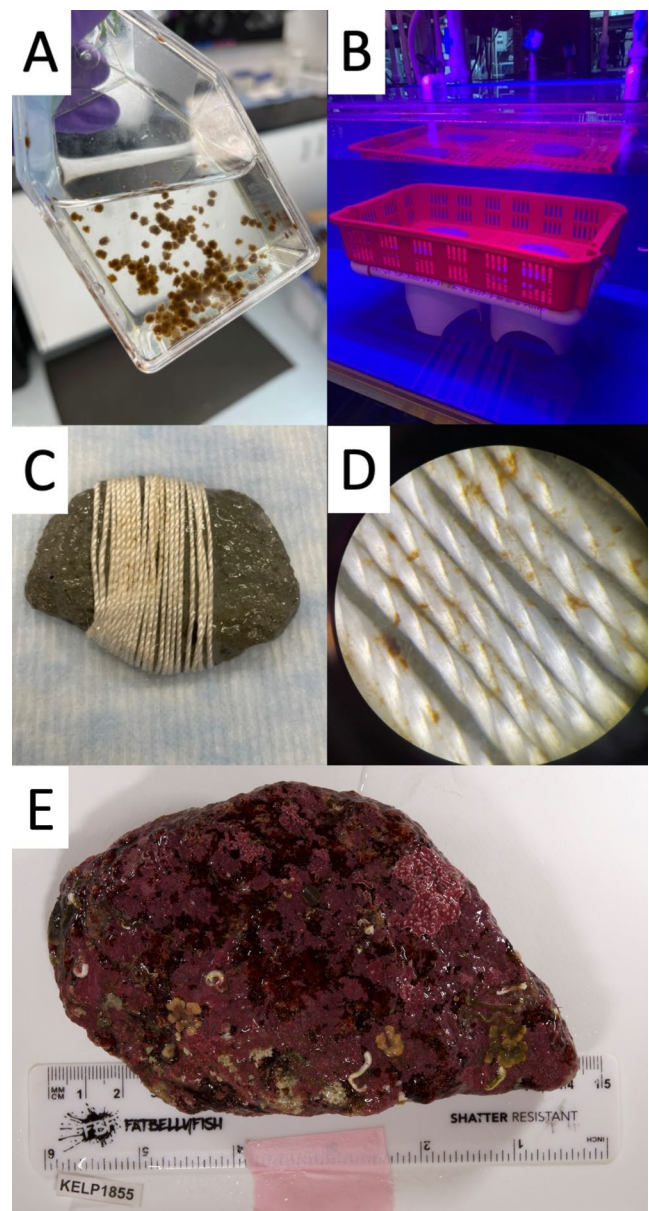


FIGURE 1 | Photograph of (A) *N.luetkeana* gametophyte clones in culture, (B) seeded gametophyte settlement rocks growing in mesocosm tanks, (C) seeded artificial gametophyte banks, (D) seeded gametophytes growing after 7 days in incubation, and (E) photograph of benthic rock sample collected in a kelp forest.

trays, and reagent reservoirs, were cleaned by soaking in a 5% bleach solution, rinsing in DI water, and treating under UV bulbs for 30 min each. Additionally, pipets were wiped with 70% ETOH and UV sterilised prior to each extraction. An additional Sterivex extraction blank and PCR blank were used as negative controls throughout sample processing.

The samples were genotyped for 11 microsatellite loci and grouped into 2 multiplexes containing 5 to 6 primer pairs (Tables S1 and S3). A PCR was run for each multiplex with 2.5 μ L of crude DNA (< 1 μ g), primer pairs in equimolar concentrations (2 μ M), 4.375 μ L of water, and 5 μ L of 1x q-solution (Qiagen Inc.) mixed with 12.5 μ L of 2x Qiagen Multiplex PCR Master Mix (Qiagen Inc.) for a single 25 μ L reaction. The programme

for multiplex PCR with eDNA is as follows: initial denaturation at 95°C for 15 min followed by 45 cycles of 94°C for 30 s, 58°C for 180 s, and 72°C for 90 s. An extension period followed for 30 min at 68°C.

2.6 | Library Preparation and High-Throughput Amplicon Sequencing

PCR products from both multiplexes were pooled in equal volumes of 10 µL (20 µL total). Pooled PCR products were cleaned using AMPure XP beads, where primers and primer dimers were removed, leaving only amplicons in each sample. The cleaned product was then uniquely barcoded in a second PCR using UD dual indexes for Illumina (IDT for Illumina DNA/RNA UD Indexes, Set A, B, C, D). The index PCR included 25 µL 2x Kapa HiFi HotStart ReadyMix (Roche Diagnostics), 10 µL water, and 5 µM mixed forward and reverse index primers (UDP0001—UDP0384) for each reaction. A library was then constructed by pooling all eDNA and tissue samples after the completion of the index PCR step. The pooled DNA library was cleaned with AMPure XP beads prior to quantification with an Agilent BioAnalyzer 2100 and a high-sensitivity DNA Kit. The libraries were diluted to 4 nM with 10 mM Tris (pH 8.5) and sequenced on an Illumina MiSeq sequencing platform (Illumina, San Diego, CA) with the MiSeq V3 600 bp kit (PE 2 × 300 bp) by the Great Lakes Genomics Center (University of Wisconsin-Milwaukee College of Freshwater Sciences).

2.7 | Scoring Alleles From Amplicon Sequences Using Amplicomsat

Reads from the MiSeq were demultiplexed and stored as a forward and reverse read for each sample. Each paired-end read was merged using FLASH (Magoč and Salzberg 2011) and exported as one file for every sample. The merged sequences were cleaned, and alleles were scored using the Amplicomsat R package. The data pipeline in Amplicomsat first scans sequences to filter reads with a quality phred score equal to or greater than 30. Reads are kept if they contain primers and n repeats of the microsatellite motif (two by default). The algorithm to score microsatellite genotypes relies on characterising the read distribution pattern for a single allele (or homozygous sample in a diploid organism). Amplicomsat can process microsatellite multilocus genotypes for individual or mixed samples (i.e., more alleles than expected from ploidy). Both run modes can score alleles by fragment size or by sequence variation, the latter revealing additional alleles which would otherwise appear to be similar by size. More details on Amplicomsat are fully described in GitHub (github.com/UWMAAlberto-Lab/Amplicomsat).

The experimentally produced eDNA samples were scored for multiple alleles at the 11 loci described above using Amplicomsat. The reference population used was Owen Bay, where 40 sporophytes (2n) were sampled and genotyped at the same loci. The PCR reagent concentrations were identical to those indicated above. The eDNA PCR conditions were adjusted to 40 cycles and a shorter 90-s

annealing time. For the reference population and eDNA samples, Amplicomsat first scored fragment size-based alleles, allowing for multiple alleles to be scored for the same locus in one sample, and then used the resulting data to search for additional sequence-based variants. Finally, the OAC for fragment size and sequence-variant alleles were counted in each sample with Amplicomsat. The genotype table for the reference population and the OAC tables for eDNA samples were used in GenotypeQuant to estimate the gametophyte numbers on the settlement rocks. These estimated NOCs were then compared to the expected value of eight gametophytes seeded on each artificial gametophyte bank.

2.8 | Estimating Gametophyte Densities in Nature

Finally, we applied these methods to eDNA samples collected in the wild to estimate natural densities of bull kelp gametophytes. During July of 2022, we collected benthic rock samples between 50 and 250 cm² ($n = 15$; Figure 1E) via SCUBA along Hakai Passage at three kelp forests, Ratatouille (51.70166, -128.08614), Breaker Group (51.736, -128.083), and Lower Islands (51.673, -128.127). Each kelp site consisted of either continuous bedrock or boulder rocky reef habitat, with very little sedimentation on the rock surfaces (Pontier et al. 2022). Epiphyte cover (e.g., coralline algae cover) varied across all collected rocks in order to represent different types of microhabitats within each site. Before sampling, each rock was photographed (Figure 1E), and the sampling area was measured using ImageJ photo analysis software (Schneider et al. 2012). The rocks were then processed for eDNA using the techniques mentioned above. Two controls were processed to verify that our sampling technique was adequate for detecting microscopic kelp growing on the benthos. Our first control ($n = 3$) was to test the presence of kelp detritus in the sterilised seawater used to rinse our rock samples. The sterile seawater was filtered with a 0.22 µm Sterivex DNA filter and fixed with Longmire's solution. The second control ($n = 3$) tested the effectiveness of our sterile rinse step conducted prior to scrubbing gametophyte DNA from the rocks. Terrestrial rocks were harvested from areas not exposed to seawater and then dipped in raw seawater (with kelp detritus) collected from the bay. The rocks were then sprayed clean with sterile seawater and processed using the above-described filtering methods. DNA was extracted from each sample and multiplex PCRs used the same techniques described for processing the seeded rocks. Microsatellite amplification products were sequenced and processed by Amplicomsat similarly to above. The resulting OAC tables were used in GenotypeQuant to estimate the NOC gametophytes from each eDNA sample. The reference population allele frequencies were estimated by genotyping at the same loci sporophyte samples from 12 neighbouring sites in Hakai Passage (51.72, -128.08) British Columbia, Canada. We collected 32 individuals from each site and pooled all the samples ($n = 384$), as their genetic structure was negligible ($F_{st} = 0.02$). We tested if GenotypeQuant estimates of NOC scaled with eDNA sampled area, using a simple linear regression in R. A Kruskal-Wallis rank sum test was performed to investigate variation between collection sites. Additionally, we extrapolated gametophyte density to a 1 m² area of optimal kelp habitat for each site by dividing sample NOC by the sampling area (m²).

3 | Results

3.1 | Testing PCR Conditions

The scoring bias of alleles from artificial eDNA with differing AR (2–5 alleles) and evenness (0.1–1) levels varied between 0.07 and 0.5 alleles when AR was held at 2, 4 and 5 alleles respectively (Figure 2A). Scoring bias varied between 0 and 0.5 alleles when allele evenness was held between 0.1 and 1 (Figure 2B).

3.2 | NMLE Precision and Accuracy

Estimated NMLE precision with a NOC of 5 had a 95% CI width between 0 and 10 when the number of alleles in a single locus varied from 2 to 25, and the number of loci varied between 10 and 40 (Figure 3A). Given the same AR and loci numbers, the 95% CI width increased and ranged between 1 and 35 contributors when the NOC was set to 15 (Figure 3B). The width of the 95% confidence interval continued to increase when the NOC was set higher to 30 and 45 gametophytes, ranging between 10 and 50 contributors (Figure 3C,D).

When testing the accuracy of the NOC estimate as a function of reference population size, both the NMLE in GenotypeQuant and methods published by Andres et al. (2023) performed similarly (Figure 4A–F). NOC estimates were negatively biased with small reference population sizes ($n=25$, Figure 4A). Methods correcting for saturated loci were more severely affected above an NOC of 15, while non-corrected methods maintained a shallower increase in bias with increasing NOC. For larger population sizes of 40

and 100, bias was low below an NOC of 20. Generally, bias increased after an NOC of 20 and shifted from negative to positive with increasing population size. Non-corrected methods always produced higher NOC estimates than corrected ones, which resulted in lower bias for smaller reference population sizes (Figure 4A,B). Precision, measured as the standard deviation of the NOC estimate among replicates, was below 5 for all NOC below 20 (Figure 4D–F). Differences in precision among all methods were negligible below an NOC of 20 (Figure 4D–F), with corrected methods being more precise for higher NOC, particularly for smaller population sizes (Figure 4D).

3.3 | HTS Amplicon Sequencing With Microsatellite Markers

Sequencing of the reference populations, laboratory eDNA, and field eDNA samples resulted in 49,288,737 reads. After demultiplexing and trimming adapters, 41,734,639 reads remained. Both reference populations sequenced from kelp tissue yielded a large number of reads per sample (mean = 56,743, SD = 17,705) compared to the field eDNA samples (mean = 38,390, SD = 13,823; Figure S1). eDNA from seeding artificial gametophyte banks had the highest number of reads per sample compared to the kelp tissue and field eDNA (mean = 83,404, SD = 24,351; Figure S1). After filtering for reads containing primers and microsatellite tandem repeat motif, tissue samples with fewer than 100 reads were excluded from analyses. An additional coverage filter was used to exclude alleles in eDNA samples with fewer than ten reads. eDNA samples collected in the laboratory trials had approximately 2x higher read numbers than field samples (Figure S1).

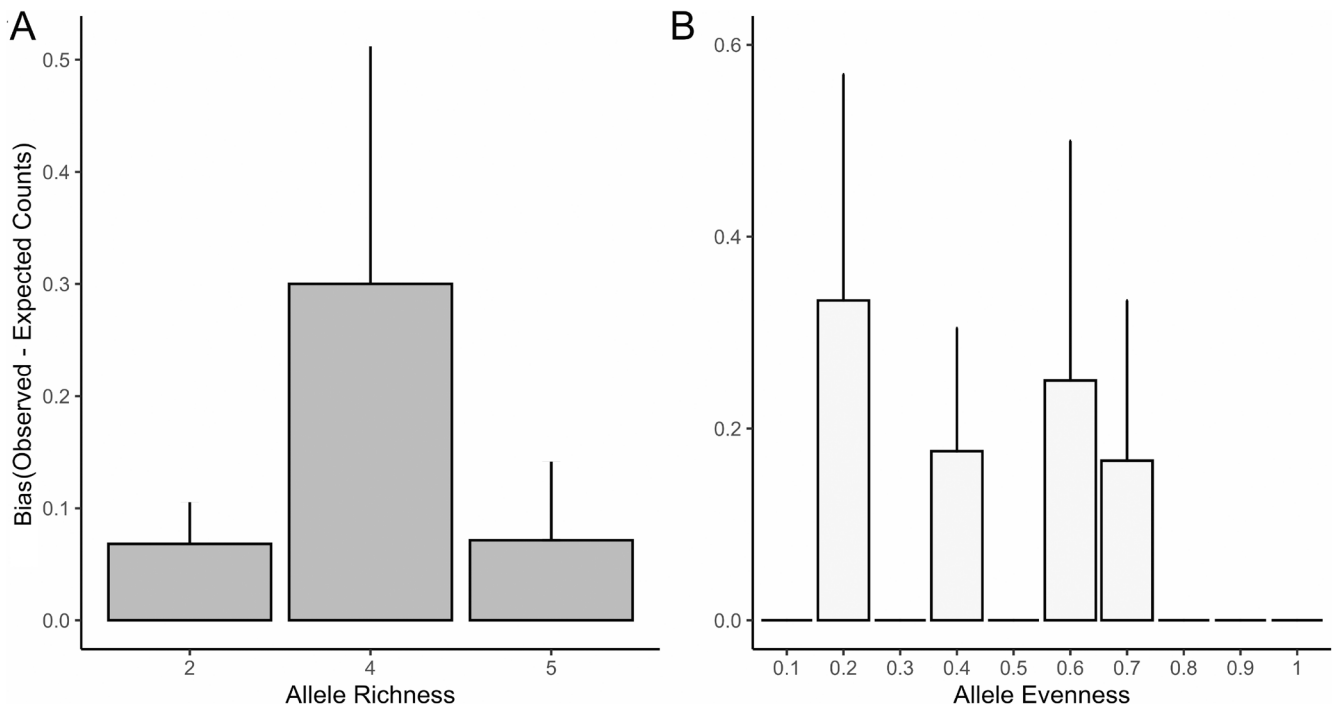


FIGURE 2 | Bar plot with standard error bars of mean differences between observed and expected allele counts from mixed samples of *N.luetkeana* gametophytes. Samples varied in (A) allele richness and (B) allele evenness.

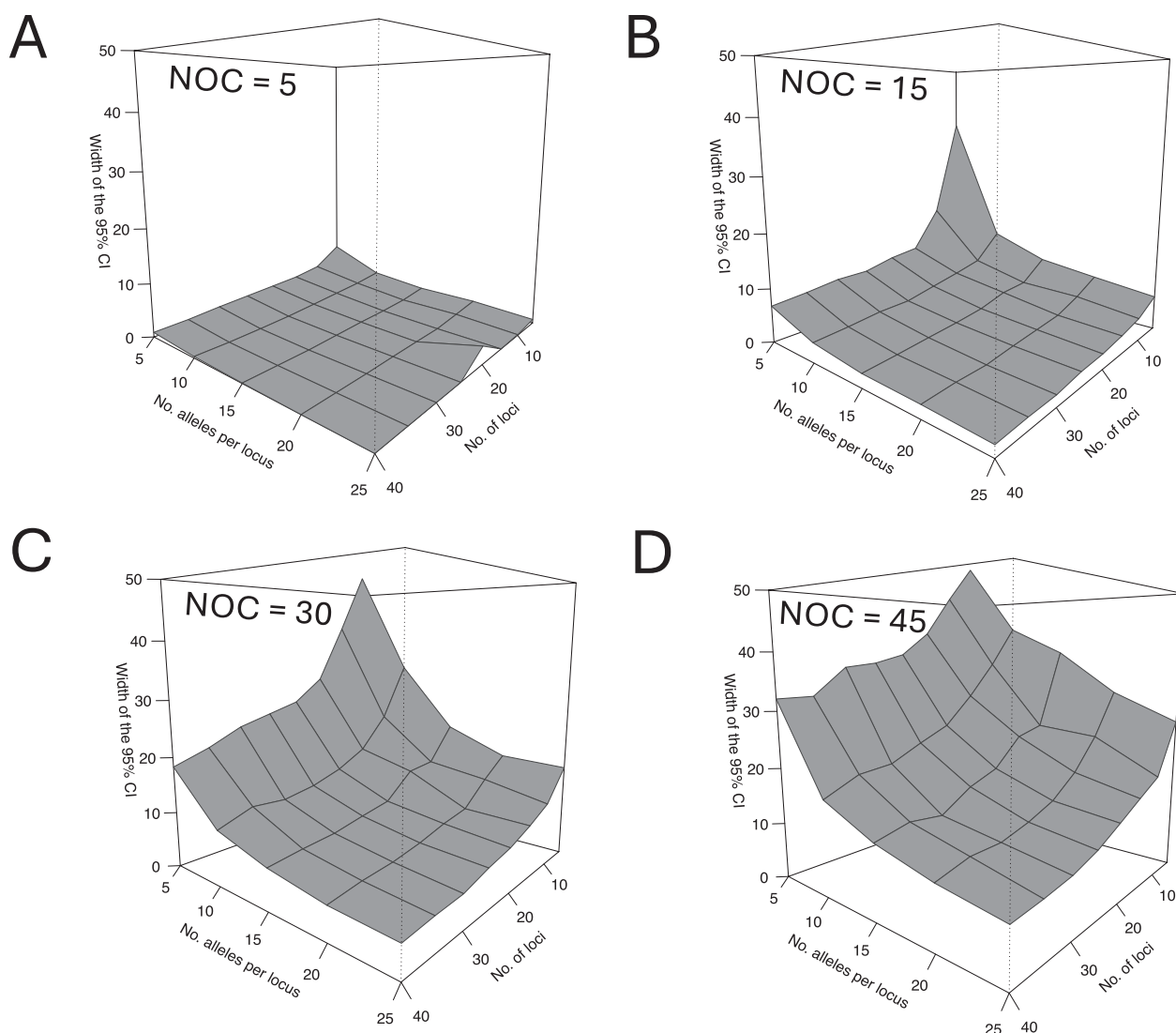


FIGURE 3 | The surface plots summarise the results of a simulation study on the effect of loci number (z-axis), allele richness (x-axis), and number of contributors in sample (NOC) on GenotypeQuant's estimate precision (width of the 95% confidence interval; y-axis). Panels A through D show four illustrative cases of simulated, known NOC (5, 15, 30, and 45).

Both the seawater and rinsing negative controls did not have sequencing reads for any of the 11 targeted loci, suggesting that all alleles sequenced from eDNA were collected directly from the rock sample and not contaminated by kelp detritus in the environment. Microsatellite markers used in this study were polymorphic, ranging between 8 and 30 fragment lengths and 11 to 65 sequence-variant alleles per locus (Table S1).

3.4 | NMLE Estimations From Laboratory Seeding Trials

Amplicon sequencing failed for 2 of the seeded artificial gametophyte-bank samples, leaving 14 samples to validate the NMLE method in GenotypeQuant. The NMLE estimated within $+3/-2$ contributors from the 8 seeded gametophytes when using OACs produced from fragment-length alleles (Figure 5A). Additionally, GenotypeQuant estimated within $+1/-2$ contributors when using OACs produced from sequence-variant alleles (Figure 5B). NOC estimation bias was similar when using

fragment-length or sequence-based alleles ($t=2.11$, $df=13$, p -value=0.055; Figure 5A,B, Table 1A,B). The precision of the NMLE to estimate NOC was higher when using sequence-variant alleles ($t=4.77$, $df=13$, p -value < 0.001; Figure 5B, Table 1B).

GenotypeQuant chi-squared tests, which flags for loci with observed allele counts deviating from their expected number of alleles given the OAC of all loci, were not significant for both allele scoring methods (Table 1A,B).

3.5 | NMLE Estimations From Field Samples

The sequencing of two markers (NL40 and Ner2) failed for most eDNA collected in the field. As a result, OACs were produced for nine markers and used to estimate NOCs with GenotypeQuant. Gametophyte NOCs estimated with OACs produced by sequence-variant alleles increased linearly from 1 to 11 individuals as the sampling area increased from 50 to 250 cm² (Figure 6A, Table 2B; $F_{1,13}=10.59$, $p<0.01$, $R^2=0.45$,

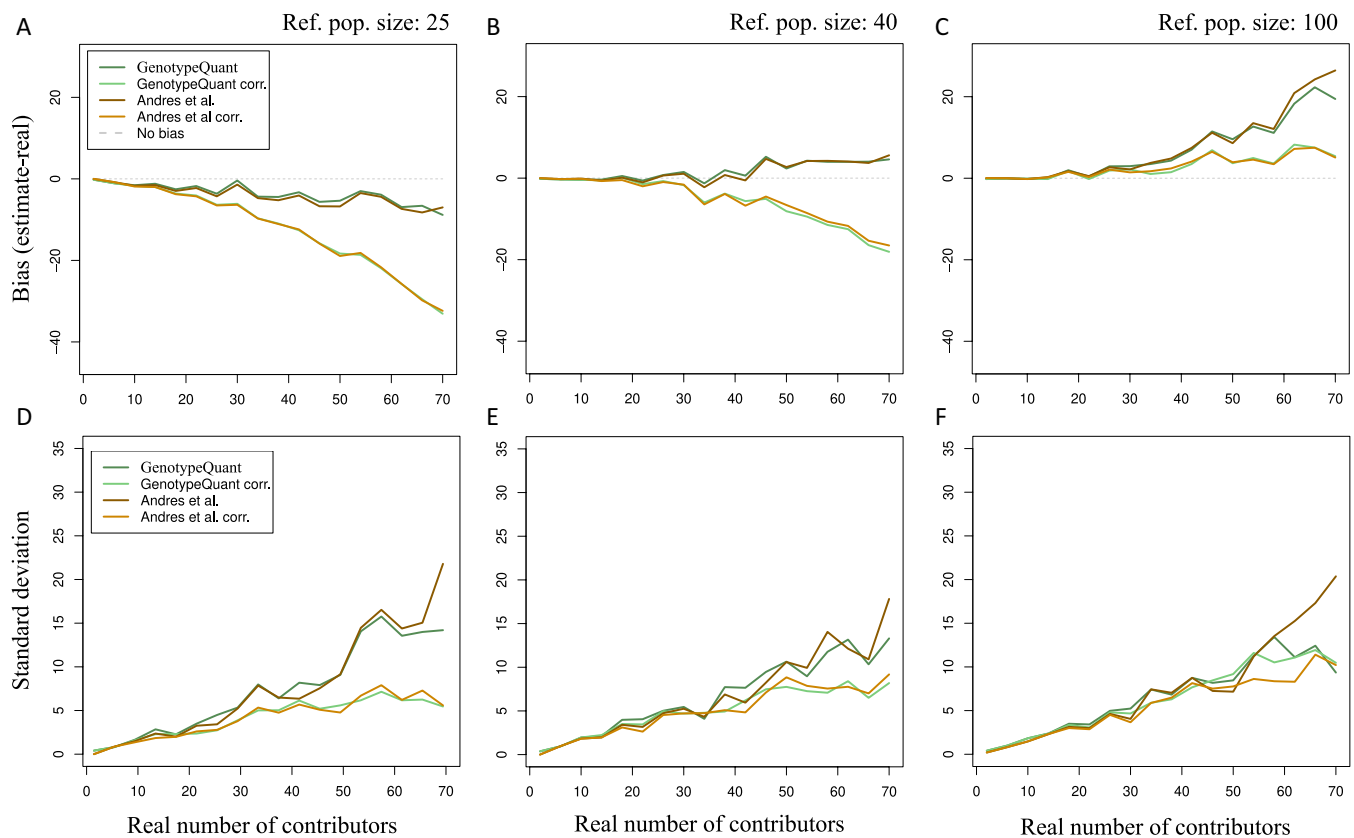


FIGURE 4 | (A, B) Line plots showing the relationship between estimation bias (the difference between estimated and real NOC) and the actual number of genotypes (simulated) in a sample. (D–F) Line plots showing the relationship between estimation precision (standard deviation over $n=25$) and the real NOC in a sample. Each plot shows trends for the corrected (removal of saturated loci) and uncorrected method used in Andres et al. 2023 (blue – uncorrected, green corrected) and the NMLE (brown – uncorrected, orange – corrected). Reference population sizes vary from (A, D) 25 individuals, (B, E) 40 individuals, and (C, F) 100 individuals.

$y=0.034x+2.159$). The relationship between NOC estimates and sampling area was not significant when populating GenotypeQuant with fragment-length OACs (Table 2A; $F_{1,13}=2.908$, $p=0.11$, $R^2=0.183$, $y=0.014x+3.103$). Both fragment-length and sequence-variant alleles had three samples (KELP1765, KELP1773, KELP1811) with significant Chi-squared values (Table 2A,B). Additionally, the Kruskal-Wallis rank sum test to identify variation between the three kelp forest sites was not significant (Chi-squared=0.091, $df=2$, $p=0.96$; Figure 6B). Average estimated total gametophyte densities per m^2 for Breaker, Lower Islands, and Ratatouille were 579, 468, and 800 respectively (Figure 6B).

4 | Discussion

This study demonstrates the successful use of molecular monitoring techniques (in the field and lab) and the parallel development of bioinformatic tools (R packages) to quantify the density of microscopic life forms, building on previous eDNA mixture analysis methods. We developed a numerical maximum likelihood estimator to estimate the number of contributors most likely to have produced the observed counts of unique microsatellite alleles. The method is akin to forensic mixture analyses, where multiple suspects are identified from contaminated DNA (Weir et al. 1997; Curran et al. 1999; Egeland et al. 2003; Grgicak et al. 2020). Using eDNA from artificial gametophyte

banks with a known genetic mix of gametophytes and eDNA collections from the field, we demonstrate the effectiveness of GenotypeQuant in estimating NOCs. These methods highlight the potential of utilising eDNA to monitor cryptic life-stages, a previously impractical task.

Using eDNA as a non-destructive monitoring tool can increase our capacity for monitoring population dynamics of species that are difficult to count with conventional field monitoring practices. Specifically, advances in next-generation sequencing provide higher resolution datasets that can capture rare alleles, rendering more accurate population allele frequencies (Lepais et al. 2020). Our study identified that AR nearly doubled 5 out of the 11 microsatellite markers when using sequence-variant alleles. Additionally, estimation precision increases when loci have more alleles, which was evident when the sequence-variant alleles counted in both our laboratory and field eDNA yielded gametophyte estimates with lower 95% confidence interval ranges.

Initial tests using a single microsatellite marker revealed that PCR could amplify mixed DNA regardless of AR and evenness. These results suggest that PCR conditions can amplify all alleles in an eDNA sample when individuals contribute different DNA concentrations. However, field eDNA sequencing data failed to evenly amplify all 11 markers in a sample. Specifically, eDNA collected from wild kelp populations had failed to sequence two

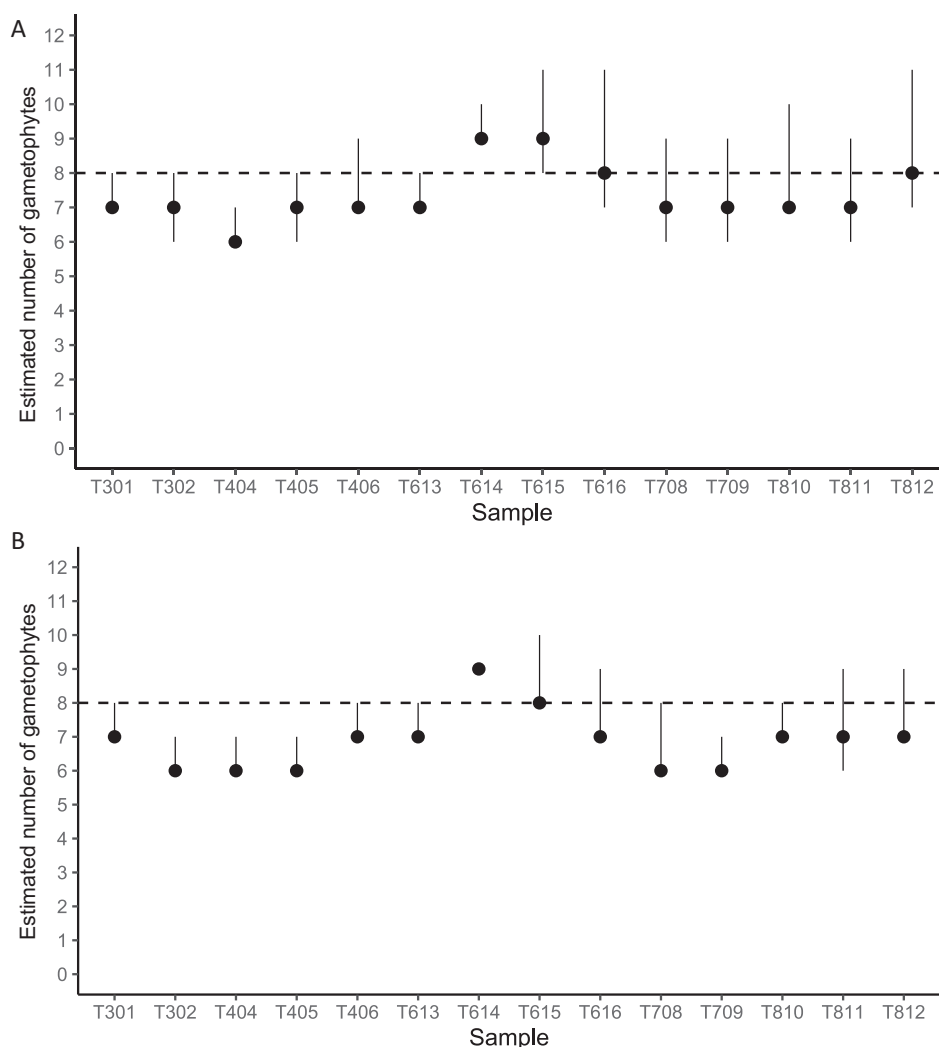


FIGURE 5 | GenotypeQuant estimates for the number of *N.luetkenana* gametophytes on artificial gametophyte banks seeded with 8 different gametophyte clones (dotted line). (Panel A) shows estimates using fragment-length allele counts, while (B) uses sequence-variant allele counts. Error bars represent the 95% confidence interval range estimated by GenotypeQuant. The x-axis shows the 14 different samples used.

or more markers in samples with lower DNA yields and, thus, lower sequencing coverage. Though amplicon HTS is a reasonable approach to analyse eDNA, our data highlight general challenges during PCR amplification that need to be considered. The presence of null alleles and allelic dropout are namely a few examples. Polymerase errors may inflate marker variation, which leads to false alleles, and ultimately biases genetic diversity estimates (McKelvey and Schwartz 2004), and in our case, NOC estimates. Furthermore, the likelihood of these amplification errors occurring increases when using eDNA due to its lower quality than DNA extracted from tissue (Monge et al. 2020). Additional optimization and troubleshooting with library preparation must be done with experimental and field samples to understand these biases further. In a subsequent study, we found that an extra cleaning and normalisation step before pooling the PCR product greatly increased sequence quality. Specifically, we cleaned the PCR product to only have amplicons within our target template range (200–400 bp). This cleaning step resulted in higher template yields after the index PCR and increased the quality (Q30) of our sequences across all markers amplified in our field eDNA samples.

NMLE estimation accuracy depended on the size of the reference population used and how fast loci become ‘saturated’, i.e., when the OAC of a locus is equal to the count of all its alleles in the reference population. When loci saturate, the NOC estimate will be positively biased, as first mentioned by Egeland et al. (2003) for diallelic markers (SNPs) and observed by simulation for multi-allelic microsatellites by Andres et al. (2023). This positive bias is offset by the effect of population size insofar as small population sizes result in fewer alleles sampled, creating a negative bias. Our simulation showed that these two sources of bias balance each other and that overall bias can be relatively small when using only 10 loci and population size 40, commonly found in microsatellite data sets. Using this level of genotyping effort, our simulation showed that methods without saturation correction-controlled bias performed better across the range of NOC explored. However, below a NOC of 20, all methods had a similar low bias. Precision was also similar across methods when NOC was below 20, while for higher NOC precision decreased faster for non-corrected methods, but only for the smallest population size of 25. When applying these MLE approaches using marker panels with lower levels of allelic richness, the

TABLE 1 | Observed allele counts (OAC) for 11 microsatellite markers and corresponding maximum likelihood (ML) estimates from seeding eight *N. luetkeana* gametophyte cultures onto artificial rocks. (A) Shows NOC estimates using fragment-length alleles. (B) Shows NOC estimates using sequence-variant alleles.

Sample	Marker												Results					
	NL21	NL24	NL25	NL26	NL27	NL30	NL40	NL47	NL50	NL59	Ner2	G.hat ^a	Lower.95. CI ^b		Upper.95. CI ^b	ChiSq ^c	p	G.Exp ^d
(A)																		
T301	7	2	2	1	4	1	1	3	2	1	1	7	7	8	10.61	0.30	8	
T302	6	2	2	1	4	1	2	4	3	2	1	7	6	8	10.02	0.35	8	
T404	6	2	1	1	3	1	NA	3	1	1	1	6	6	7	9.16	0.33	8	
T405	6	3	2	1	4	1	2	4	3	1	1	7	6	8	9.31	0.41	8	
T406	7	4	2	1	4	1	1	4	2	2	1	7	7	9	11.62	0.24	8	
T613	7	3	2	1	3	1	1	3	2	1	1	7	7	8	9.56	0.39	8	
T614	9	3	2	1	4	1	1	3	2	2	1	9	9	10	11.26	0.26	8	
T615	8	4	2	1	4	1	2	4	3	2	1	9	8	11	10.30	0.33	8	
T616	7	3	2	1	4	1	NA	4	3	2	1	8	7	11	8.15	0.42	8	
T708	5	2	1	1	2	1	1	1	2	1	NA	7	6	9	8.19	0.41	8	
T709	6	3	2	1	4	1	2	4	3	2	1	7	6	9	9.53	0.39	8	
T810	7	3	2	1	4	1	NA	4	2	2	1	7	7	10	8.38	0.40	8	
T811	6	3	2	2	4	1	2	4	3	2	1	7	6	9	8.12	0.52	8	
T812	7	3	2	2	4	1	NA	4	3	2	1	8	7	11	6.69	0.57	8	
(B)																		
T301	7	2	2	1	4	1	1	3	3	1	1	7	7	8	8.52	0.48	8	
T302	6	2	2	1	4	1	2	4	4	2	1	6	6	7	9.90	0.36	8	
T404	6	2	1	1	3	1	NA	3	2	1	4	6	6	7	13.52	0.10	8	
T405	6	3	2	1	4	1	2	4	4	1	1	6	6	7	10.21	0.33	8	
T406	7	4	2	1	4	1	1	4	3	2	1	7	7	8	10.77	0.29	8	
T613	7	3	2	1	3	1	1	3	3	1	1	7	7	8	7.51	0.58	8	
T614	9	3	2	1	4	1	1	3	2	2	1	9	9	9	9.47	0.40	8	
T615	8	4	2	1	4	1	2	4	4	2	1	8	8	10	9.73	0.37	8	

(Continues)

TABLE 1 | (Continued)

Sample	Marker												Results				
	NL21	NL24	NL25	NL26	NL27	NL30	NL40	NL47	NL50	NL59	Ner2	G.hat ^a	Lower.95. CI ^b	Upper.95. CI ^b	ChiSq ^c	p	G.Exp ^d
T616	7	3	2	1	4	1	NA	4	4	2	2	7	7	9	8.57	0.38	8
T708	5	2	1	1	2	1	1	1	2	1	NA	6	6	8	8.98	0.34	8
T709	6	3	2	1	4	1	2	4	4	2	1	6	6	7	9.46	0.40	8
T810	7	3	2	1	4	1	NA	4	3	2	1	7	7	8	8.74	0.36	8
T811	6	3	2	2	4	1	2	4	5	2	2	7	6	9	8.59	0.46	8
T812	7	3	2	2	4	1	NA	4	4	2	2	7	7	9	7.08	0.53	8

^aMaximum likelihood estimate from the NMLE.
^b95% confidence interval range (upper and lower values).
^cChi-square statistic.
^dExpected number of seeded gametophytes.

user would need to increase their marker panel to maintain similar precision estimates presented in this study.

Given the many markers required for reasonable precision and accuracy, no MLE approach is practical for estimating large NOC values. To tackle potentially large NOC values, pilot data should be collected to decide how the captured volume or sampling area can be reduced to control the magnitude of NOC. With our kelp gametophytes, we did not count or estimate any NOCs larger than 13 contributors in an eDNA sample, and therefore we expect bias to be small when using the correction and smaller reference population sizes. It is also important to consider the population history of the reference population used in the study, as the NMLE depends on the population allele frequencies to estimate NOCs and the Chi-square statistic. If the effective population size drastically changes between sampling for the reference pool of allele frequencies and eDNA collections, the NMLE would not accurately estimate the true NOC. However, in one study with giant kelp, we showed that allele frequencies were stable over a period of 10 years (Klingbeil III et al. 2022).

Although this study further builds the potential of eDNA to estimate species' densities, there were problems with obtaining sequences from every microsatellite marker in the panel. Difficulties arose when sequencing eDNA collected from the field, where all 15 samples failed to have sequences for every marker, including two markers that failed to sequence entirely. The sample's quality might be affected because the DNA is not harvested directly from an organism but rather collected indirectly from the environment. Primarily, the sample is mixed with an array of different types (and concentrations) of non-target DNA, which can prohibit the markers from finding the target DNA during PCR (Adams et al. 2019). Additionally, eDNA tends to be highly fragmented, which could lead to a loss of marker template in the sample, resulting in field samples not containing DNA associated with the microsatellite marker, a complication that was mentioned in previous studies. To combat the caveats associated with sequencing microsatellites from eDNA, GenotypeQuant produces an output report associated with each NOC estimate. This report includes a Chi-square estimate that compares observed allele counts at each locus to an expected number derived from the reference population's allele frequencies and the OAC measured across all loci. The Chi-square can be interpreted as a measure of the OAC accuracy for each locus due to allele dropout or eDNA quality. In the context of counting gametophyte NOCs from the field, three samples had significant Chi-square estimates with fragment-length and sequence-variant OACs, where PCR only amplified 2 out of the 11 markers. As such, the Chi-square value allows users to parse out problematic NOC estimates or to exclude the problematic loci, flagged by the Chi-square statistics, from the NMLE estimation.

The analysis of field eDNA samples showed that NOC scaled with the sampling area of the boulders analysed. This association is reassuring that NOC estimates produced by the NMLE represent microscopic kelp densities in wild populations. Our field collections occurred in July of 2022, just before the spores of that year were released (Pontier et al. 2022). Therefore, the microscopic kelp estimated in this study was produced at least in the previous year and belongs to the same cohort of the adult

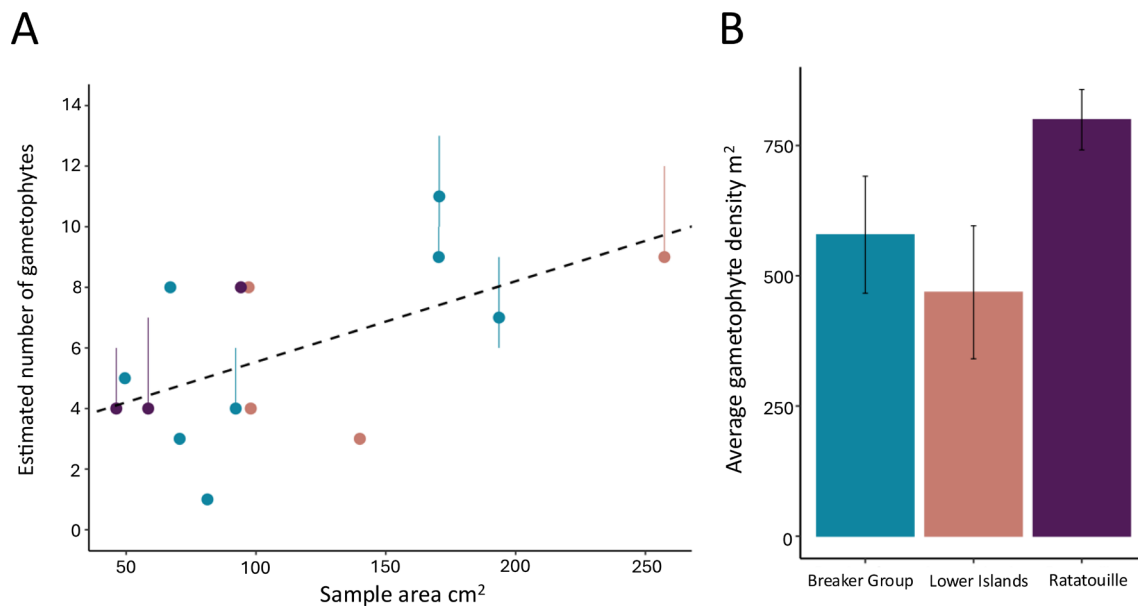


FIGURE 6 | (A) Linear association between the GenotypeQuant's estimated number of *N. luetkeana* gametophytes (from sequence-variant allele counts) and sampling area (cm²). Samples were collected from 3 kelp forest sites along Hakai Pass in British Columbia, Canada (Ratatouille, Breaker Group, and Lower Islands). Error bars represent the 95% confidence interval range estimated by GenotypeQuant. (B) Extrapolated gametophyte numbers per m². Error bars represent the standard error of density estimates for each site.

sporophytes in the collection year. The extrapolated ~500 haploid genotypes·m⁻² density, assuming optimal benthic habitat, was much higher than the densities of adult sporophytes (~3–6 sporophytes·m⁻²; Pontier et al. 2022). If the genotypes counted were microscopic sporophytes that delayed their recruitment, their higher density to the adult sporophyte population is expected, given self-thinning in kelp populations (Dean et al. 1989). In our gametophyte culture work, we have observed that keeping sporophytes viable at small sizes is much more complicated than gametophytes. It is also difficult to explain why sporophytes of the same cohort would arrest their development at the microscopic stage while others developed into adults. If indeed we estimated gametophytes from the previous year, their density was much lower than 1×10^6 gametophyte m² required for successful fertilisation (Reed et al. 1991). However, in late winter, when the year's cohort of sporophytes was recruited, gametophyte density was not limiting of fertilisation, and it is likely to decrease in the next 5–6 months. In our gametophyte culture work, a non-negligible proportion of gametophytes do not produce sporophytes when sexual reproduction is induced despite all gametophytes looking equally healthy and continuing to grow. With only the work presented here, we cannot conclude if gametophytes present in July will be viable and contribute to next year's generation; in this case bull kelp would have overlapping gametophyte generations and an actual “seed” bank. Future studies should temporally monitor gametophytes via eDNA sampling and compare seasons with and without active spore release to quantify gametophyte depletion in the winter and accumulation from early summer to late fall.

Regarding gametophyte density required for fertilisation, the NMLE approach estimates genotype densities, which could be much lower than those of gametophyte vegetative fragments—if gametophytes fragment in the wild as they do in culture. In culture conditions, gametophytes vegetatively propagate

and fragment cells as they grow (Lüning and Neushul 1978; Destombe and Oppliger 2011; Wade et al. 2020), creating hundreds to thousands of copies of a single genotype, which led to conjecture on the role that vegetative fragmentation can play in dispersal (Kain 1979; Destombe and Oppliger 2011; Ebbing et al. 2020). In the field, kelp species tend to live in very stressful hydrodynamic environments, including areas that experience constant wave forces, tidal currents, or both (Koehl and Wainwright 1977; Martone 2007; Koehl et al. 2008; Breikreutz et al. 2022). This environment could promote gametophyte fragmentation by increasing clone numbers and total gametophyte densities in wild kelp populations.

We developed two R packages to integrate all bioinformatic steps required to process sequencing reads and estimate NOC from eDNA using an NMLE approach. While other software can process amplicon-sequenced microsatellite reads for fragment size or sequence-based polymorphism and provide classical population genetics (Zhan et al. 2017; Roy et al. 2021; Liu et al. 2024), Amplicomsat is unique in additionally scoring OAC from mixed samples. To our knowledge, only one other publicly available software can provide OAC, the FDStools pipeline (Hoogenboom et al. 2017). However, FDStools depends heavily on bash scripting over several subroutines to get from processing reads to OAC and to produce diagnostic plots (Lepais et al. 2020). Other available software that falls short of providing an OAC is well summarised in Liu et al. (2024) publication describing the Seq2Sat and SatAnalyzer toolkit. Concerning the estimation of NOC, the NMLE GenotypeQuant had similar precision and accuracy to the analytical maximum likelihood method used in Andres et al. (2023).

GenotypeQuant is slower than the latter method due to the initial generation of OAC distributions from the reference population allele frequencies. However, the NMLE creates sampling

TABLE 2 | Observed allele counts (OAC) for 11 microsatellite markers and corresponding maximum likelihood (ML) estimates from rocks (between 46 and 260 cm²) collected in *N. luelkeana* forest habitats. (A) shows NOC estimates using fragment-length alleles. (B) shows NOC estimates using sequence-variant alleles.

Sample	Site	Area (cm ²)	Marker										Results				
			N121	N124	N125	N126	N127	N130	N147	N150	N159	G.hat ^a	lower.95.CI ^b	upper.95.CI ^b	ChiSq ^c	p [*]	
(A)																	
KELP1765	Breaker	67	3	2	1	NA	5	NA	3	NA	NA	5	5	5	8.75	0.03*	
KELP1768	Breaker	49.5	2	3	NA	NA	4	1	2	1	1	4	4	4	4.36	0.50	
KELP1769	Breaker	170.6	NA	4	7	NA	7	2	4	NA	2	8	7	11	8.66	0.07	
KELP1773	Breaker	193.6	NA	3	NA	3	4	NA	4	NA	3	4	4	6	3.67	0.30	
KELP1775	Breaker	92.2	NA	1	NA	NA	4	NA	3	NA	NA	4	4	6	5.60	0.02*	
KELP1777	Breaker	70.6	NA	1	NA	NA	3	NA	NA	NA	2	3	3	3	3.25	0.07	
KELP1778	Breaker	81.3	NA	NA	NA	NA	1	NA	1	NA	1	1	1	1	0.13	0.72	
KELP1781	Breaker	170.4	7	NA	NA	2	4	NA	2	NA	4	7	7	9	1.19	0.75	
KELP1855	Lower Islands	98	NA	2	4	NA	1	NA	1	NA	1	4	4	4	2.80	0.42	
KELP1857	Lower Islands	257.3	NA	3	5	NA	6	NA	2	NA	NA	6	6	8	4.96	0.08	
KELP1866	Lower Islands	97.2	5	7	6	1	6	4	4	NA	4	8	7	8	5.48	0.48	
KELP1870	Lower Islands	140	NA	1	NA	2	NA	NA	1	NA	3	3	3	3	2.25	0.33	
KELP1807	Ratatouille	94.2	6	NA	NA	NA	3	NA	1	NA	3	6	6	7	0.86	0.65	
KELP1811	Ratatouille	46.2	NA	NA	NA	NA	3	NA	2	NA	NA	3	3	6	0.04	<0.001*	
KELP1814	Ratatouille	58.5	NA	NA	NA	1	NA	NA	NA	3	4	4	4	5	1.86	0.17	
(B)																	
KELP1765	Breaker	67	4	2	1	NA	7	NA	4	NA	NA	7	7	7	9.82	0.02*	
KELP1768	Breaker	49.5	3	3	NA	NA	5	1	3	1	1	5	5	5	5.16	0.40	
KELP1769	Breaker	170.6	NA	5	8	NA	10	2	4	NA	2	11	10	13	6.36	0.17	
KELP1773	Breaker	193.6	NA	4	NA	5	6	NA	5	NA	3	7	6	9	2.40	0.49	
KELP1775	Breaker	92.2	NA	1	NA	NA	4	NA	4	NA	NA	4	4	6	4.67	0.03*	

(Continues)

TABLE 2 | (Continued)

Sample	Site	Area (cm ²)	Marker										Results			
			NI21	NI24	NI25	NI26	NI27	NI30	NI47	NI50	NI59	G.hat ^a	lower.95.CI ^b	upper.95.CI ^b	ChiSq ^c	p [*]
KELPI777	Breaker	70.6	NA	1	NA	NA	3	NA	NA	NA	2	3	3	3	1.72	0.19
KELPI778	Breaker	81.3	NA	NA	NA	NA	1	NA	1	NA	1	1	1	1	0.10	0.75
KELPI781	Breaker	170.4	9	NA	NA	2	7	NA	2	NA	4	10	9	11	2.61	0.46
KELPI855	Lower Islands	98	NA	2	4	NA	2	NA	1	NA	1	4	4	4	1.74	0.63
KELPI857	Lower Islands	257.3	NA	3	6	NA	9	NA	3	NA	NA	11	9	12	4.99	0.08
KELPI866	Lower Islands	97.2	6	7	7	1	8	4	5	NA	4	8	8	8	6.22	0.40
KELPI870	Lower Islands	140	NA	1	NA	2	NA	NA	2	NA	3	3	3	3	2.60	0.27
KELPI807	Ratatouille	94.2	8	NA	NA	NA	4	NA	1	NA	3	8	8	9	1.23	0.54
KELPI811	Ratatouille	46.2	NA	NA	NA	NA	4	NA	2	NA	NA	4	4	6	0.09	>0.001*
KELPI814	Ratatouille	58.5	NA	NA	NA	1	NA	NA	NA	4	4	4	4	6	2.41	0.12

^aMaximum likelihood estimate from the NMLE.
^b95% confidence interval range (upper and lower values).
^cChi-square statistic.
^{*}Significant *p*-value.

distributions of OAC for each possible value of NOC and locus, which do not require combinatorial calculations or removing markers with a high number of alleles. Thus, the computational limits presented in Andres et al. (2023) are not present when using GenotypeQuant. The generation of the OAC distributions also allows for flexibility in the sampling scheme to address deviations from Hardy–Weinberg, like the F_{IS} correction implemented in the current version, which can otherwise make a likelihood function difficult to derive.

Author Contributions

L.M.L. and F.A. conceived the study. L.M.L., F.A. and I.G. obtained project funding. O.P., K.C.R., I.G., and L.M.L. designed the study. L.M.L., K.C.R., B.R.I. and O.P. collected the data. K.C.R. and B.R.I. cultivated and maintained gametophyte cultures used in this study. F.A. developed the R packages. L.M.L. performed the analysis. I.G. and F.A. provided laboratory support and supervision for the study. L.M.L. drafted the first version of the manuscript, followed by F.A. revision. All authors revised and approved the final manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets presented in this study can be found in the following online repositories:

Raw genetic sequences are deposited in NCBI SRA (PRJNA1235304). Primer sequences are available on GenBank (BankIt2936207). R packages are available in GitHub: <https://github.com/UWMAAlberto-Lab/Amplicomsat>, <https://github.com/UWMAAlberto-Lab/GenotypeQuant>. R code used for GenotypeQuant simulations is available in GitHub: <https://github.com/UWMAAlberto-Lab/liggan.et.al.2025.simulations.git>. Reference population genotypes and gametophyte culture genotypes are available in Dryad (doi:10.5061/dryad.r4xgxd2q6).

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.